Image Segmentation for Radar Signal Deinterleaving Using Deep Learning

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Passive systems, such as electronic intelligence and electronic support measures systems, aim to extract necessary information from the received radar signals for situational awareness. To achieve this, the system must first deinterleave the radar signals simultaneously coming from different emitters, so that the pulse repetition interval (PRI) patterns will be revealed for further analysis and identification purposes. PRI transform is a well-known deinterleaving method that utilizes the complex autocorrelation function. There are two main versions of the method. The initial version detects only constant PRI schemes, while the second modified version is capable of detecting varying PRI schemes as well. Miss detection of varying PRI patterns is the drawback for the first version, while producing harmonics, especially at high PRI levels, is the disadvantage of the second one. To alleviate these problems, we propose an image segmentation method based on deep learning. The developed preprocessing step uses both versions of the PRI transform outputs to generate 2-D time–PRI images of the collected radar emissions, so that constant and varying PRI patterns are revealed. The images are concatenated and fed to the proposed network, which uses a practicable U-Net structure. The output of the network directly estimates the PRI levels of the existing radars and the time duration of the transmission jointly. In addition to qualitative and quantitative experiments on the synthetic datasets, qualitative experiments are conducted on real measurements, in which we demonstrate that the proposed method effectively utilizes PRI transform in the preprocessing step and outperforms both versions of the PRI transform in terms of accuracy, Jaccard index, structural similarity, and PRI estimation error metrics.

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I. INTRODUCTION

Electronic intelligence and electronic support measure (ESM) systems are passive electronic warfare (EW) systems that analyze the received radar signals without any transmission [1]. The detected pulses are encoded into pulse description words (PDWs), which contain interpulse parameters, such as time of arrival (TOA), pulse width (PW), frequency, and angle of arrival (AOA). Note that, at this point, each PDW can belong to different radar, since they are interleaved in the time domain. To identify the radars from the collected PDWs, they must be deinterleaved first [2]. When the signals coming from different emitters are well-separated in PW, frequency, or AOA domains, basic clustering algorithms might successfully deinterleave the collected PDWs without the TOA analysis [3]–[5]. However, as a result of the rapid developments in the fields of communication and radar systems, the frequency spectrum for a typical EW environment is getting more crowded, hence simple clustering techniques become insufficient for successful deinterleaving [6]–[8]. In addition, many EW systems are not able to measure AOA for each received pulse, and AOA information may be missing in the generated PDWs, which also affects the clustering performance adversely. Therefore, there is a need for a deinterleaving algorithm, which is capable of the TOA analysis. Our work tackles the scenarios, in which the radar signals have indistinguishable or unavailable PW, frequency, and AOA parameters, and it presents a novel TOA analysis algorithm.

The deinterleaving operation is unsupervised in the sense that there is no prior information regarding the number or the signal characteristics of the existing radars in the measurement environment [2]. PRI transform is a well-known deinterleaving technique, which is a spectral method that builds its test statistic by utilizing the complex autocorrelation function of the pairwise differences of TOAs (DTOAs) values [9]. The method is very effective for capturing constant PRI patterns, robust to missing pulses, and does not produce harmonics. However, it is not able to detect the varying pulse repetition interval (PRI) patterns, since the phases of the complex autocorrelation toggle and become destructive for the test statistic. To overcome this problem, the improved method uses overlapped PRI bins and adaptively changes the time origin for each pairwise DTOA, so that the phases match and construct a higher test statistic, which enables the detection of varying PRI patterns as well [10]. This advancement comes at a price, as the method is likely to produce harmonics and false alarms, especially at high PRI levels. Our method aims to remove the drawbacks that are individually caused by these techniques. It leverages both techniques in the preprocessing step in a way that each of their strengths is used to outweigh the other's disadvantage.

Different from the PRI transform, the deinterleaving problem can be viewed as an online processing procedure, and it can be solved by the algebraic matrix operations [11]. In this manner, the inverse of the TOA matrix of the data reveals certain PRI patterns, so that the constant PRI-type yields a tridiagonal matrix and the staggered PRI-type results in repeated elements on the diagonal of the matrix. One obvious disadvantage of this method is the computational load when the number of pulses N is large. Due to the matrix inversion, the method involves N^3 elementary operations. Note that today's ESM receivers can generate one million pulses per second. Moreover, in another work, the PRI patterns are illustrated with regular grammars, and finite automata are developed to realize online pulse deinterleaving [12]. The states of the pulse patterns are stored and continuously updated in the automata, and currently received pulses are analyzed through the associated finite-state controls to determine their belongings to a certain emitter. While constructing the automata, the method has concrete assumptions on the stagger level and state number of the emitters, which degrades its utility. The main drawbacks of these online deinterleaving methods are that they only work for repetitive sequences, such as constant and staggered PRI-types. Therefore, they do not cover the real problem scenario, in which the radars with jittered PRI-type are also observed.

Lately, with the advancements in the hardware technology, artificial intelligence takes part in almost every field, including EW systems. The effectiveness of neural networks for EW problems that cannot be mathematically modeled and optimally solved have been demonstrated in various papers [13]–[17]. Most of these works utilize the features that are extracted from the raw data in the learning process. The features can be obtained in two ways: 1) the first is the feature learning approach that automatically discovers the representations needed; and 2) the second is the manual technique that involves expert knowledge. While the automatic technique has the end-to-end learning opportunity that brings the computational advantage, the manual technique has the advantage of benefiting from the prior domain knowledge [18]–[21]. Getting motivated by these works, we propose a new method, which benefits from feature extraction and utilizes deep neural networks for the radar pulse deinterleaving problem. In our method, first we transform the deinterleaving problem into an image segmentation problem by the developed preprocessing step. The preprocessing step manually extracts the necessary features, and it applies both versions of the PRI transforms to the given data, so that two different time–PRI images are formed. These images are then concatenated and fed to a proposed deep neural network that can solve the image segmentation problem. There are numerous image segmentation models based on convolutional neural networks, autoencoders, and attention mechanisms [22]–[24]. U-Net utilizes both downsampling and upsampling operations to capture different levels of features, and it yields state-of-the-art segmentation performance [25]. Several other versions of its aim to enhance its segmentation performance by making changes in the design of the network [26]–[28]. In our work, we design a practicable U-Net structure that has much fewer parameters than that of the original work. This enables faster inference and removes the issue of overfitting. The network makes pixelwise classification by learning to fit the concatenated input image to the given label image indicating the class of each pixel. Finally, the proposed network jointly estimates the center PRI levels of the existing radars and the time duration of their transmission. Therefore, tracking the deinterleaved pulses, which is the procedure followed after the deinterleaving operation, becomes much easier. In addition to qualitative and quantitative experiments, which are performed on synthetic data, we conduct qualitative experiments on real data and demonstrate that the proposed method utilizing PRI transform in the preprocessing step outperforms both versions of the PRI transforms in terms of accuracy, Jaccard index, structural similarity (SSIM), and PRI estimation error metrics. To the best of our knowledge, our work is the first attempt to solve the radar signal deinterleaving problem by approaching from the perspective of cognitive image segmentation, and it is the first work that shares results on real measurements.

To sum up, the main contributions of this article are as follows.

1) We present a novel preprocessing method that applies PRI transform to generate 2-D time–PRI images that are the inputs of the proposed deep neural network.
2) We develop a deep neural network that achieves pixelwise classification on the generated PRI transform images.
3) We share qualitative and quantitative results on synthetic and real data, which validates the success of our work.

The rest of this article is organized as follows. Section II gives some preliminaries about the PRI types and PRI transform. Section III shares the details of the proposed method. Section IV provides simulation results and comparisons. Finally, Section V concludes this article.

II. PRELIMINARIES

In this section, we introduce PRI types and provide a brief review of the PRI transform, which is the basic building block of our preprocessing stage.

A. PRI Types

Most of the radars follow similar PRI patterns to achieve their objectives. These patterns belong to four main classes: 1) constant; 2) jittered; 3) staggered; and 4) dwell switch. Constant PRI means the usage of the same PRI, while jittered PRI corresponds to highly varying PRI schemes. Staggered PRI is the periodic PRI pattern, and dwell switch

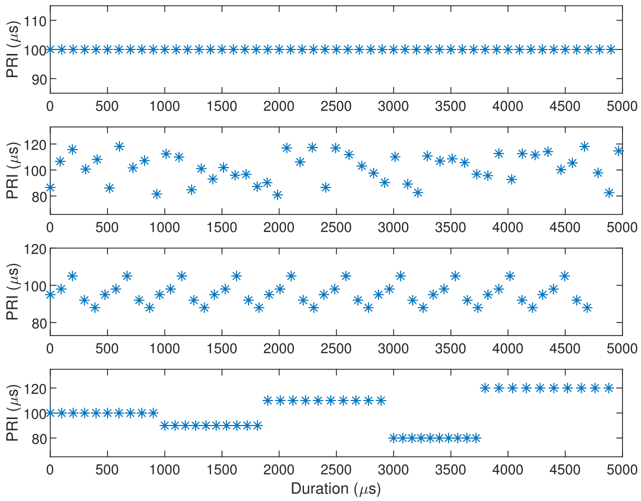

Fig. 1. PRI types. Constant, jittered, staggered, and dwell switch PRI patterns are shown from top to bottom in the subplots, respectively.

PRI is the usage of several constant PRI schemes sequentially. Fig. 1 shows the PRI patterns when there is no missing pulse or erroneously detected nonexisting pulses.

B. PRI Transform

The PRI transform has two versions: 1) the first version, PRI transform-I (PRIT-I), is very powerful for detecting constant PRI patterns and does not generate harmonic PRIs, i.e., integer multiples of the true PRI value [9]; and 2) the second one, PRI transform-II (PRIT-II), is developed for varying PRI sequences [10]. We summarize these two methods as follows.

1) *PRI Transform-I [9]:* Let N be the number of pulses. The arrival times of the intercepted pulses can be expressed as a sum of unit impulses as

$$g(t) = \sum_{n=0}^{N-1} \delta\left(t - t_n\right) \tag{1}$$

where t_n is the TOA of the nth pulse and $\delta(.)$ is the Dirac delta function. PRI transform is defined by the complex autocorrelation function of DTOA values as the following:

$$D(\tau) = \sum_{n=1}^{N-1}\sum_{m=0}^{n-1} \delta\left(\tau - t_n + t_m\right) \exp\left[\frac{2\pi i t_n}{t_n - t_m}\right]. \tag{2}$$

Note that the phase factor $\exp(2\pi i t_n/(t_{\rm n} - t_{\rm m})]$ in the complex autocorrelation suppresses the subharmonics, which would occur at high PRI levels of the autocorrelation function.

PRI transform is numerically computed in the range $[\tau_{\min}, \tau_{\max}]$, where $\tau_{\min}$ and $\tau_{\max}$ are the minimum and maximum PRI levels to be analyzed, respectively. Let b_k be the bin width of kth PRI level, then the discrete PRI transform is calculated as

$$D[k] = \sum_{\substack{(m,n)\\ \{\tau_k - b_k/2 \le t_n - t_m < \tau_k + b_k/2\}}} \exp\left[\frac{2\pi i t_n}{t_n - t_m}\right]. \tag{3}$$

2) *PRI Transform-II [10]:* PRIT-I is not able to capture varying PRI patterns, since the peak of the test statistic reduces dramatically when the PRI values are not constant. There are two reasons behind this reduction: a) the first one is the phase error of the phase factor in the complex autocorrelation function; and b) the second one is the spread of the peak power around the center PRI. The modified algorithm aims to solve these problems by shifting the time origins and using overlapped PRI bins. The time origins are shifted as follows. First, the preliminary phase is determined by

$$\eta_0 = \frac{t_n - O_k}{\tau_k} \tag{4}$$

where O_k denotes the previous time origin of the kth PRI level and $\tau_k = t_n - t_m$. Then, the phase is decomposed as

$$\eta_0 = \nu(1+\zeta) \tag{5}$$

where ν is an integer and ζ is a real number, such that $-1/2 < \zeta < 1/2$. If $(\nu = 1 \quad \wedge \quad t_m = O_k)$ or $(\nu \ge 2 \quad \wedge \quad |\zeta| \le \zeta_0)$, t_n is assigned as the new time origin, where ζ_0 is a positive parameter that determines the mobility of the time origins. Furthermore, bins are overlapped to avoid the reduction of the peak in varying PRI values. The width of the kth PRI bin is then set as $b_k = (\tau_{\max} - \tau_{\min})/K$ and the center of each PRI level is calculated as

$$\tau_k = (k - 1/2)b_k + \tau_{\min}, \quad k = 1, 2, \ldots, K \tag{6}$$

where $[\tau_{\min}, \tau_{\max}]$ is the PRI range to be analyzed and K is the number of PRI levels. Finally, the following threshold is applied to the test statistic $D[k]$:

$$\max\left(\alpha\frac{T}{\tau_k}, \beta C_k, \gamma\sqrt{T\rho^2 b_k}\right) \tag{7}$$

where T is the time window duration, C_k is the total number of pairwise (m, n) pulses that obeys $\tau_k - b_k/2 < t_n - t_m < \tau_k + b_k/2$, and $\rho = \frac{N}{T}$ is the pulse density. The constants α, β, and γ are the tunable parameters, which are selected as $\alpha = 0.7$, $\beta = 0.15$, and $\gamma = 3$ in the proposed preprocessing step, where α sets the required pulse density, β determines whether τ_k is a PRI or its subharmonics, and γ eliminates the noise under the assumption that TOAs come from the Poisson distribution. In the original paper, only the parameter α is differently selected as 0.3 [10]. This value is intentionally increased in our work, since PRI transform is utilized in a shorter time process provided by the windowing operation described in the following section.

III. PROPOSED TECHNIQUE

Each version of the PRI transform has individual disadvantages and strengths. Since it is not possible to combine them analytically, we prefer to apply a preprocessing step utilizing both methods, so that we generate input images from which the proposed deep model can automatically learn.

A. Proposed Preprocessing Step

In the preprocessing step, both versions of the PRI transforms are used to generate 2-D images. X and Y axes of the images represent the time and the PRI level, respectively. In the simulation data, PRI levels are taken from [5,204] μs. Note that the values of these bounds are not important to the network, since it processes images instead of the raw TOA data. Hence, it will produce consistent results even if the bounds are changed, which will be discussed in the following section. Due to the bound assumption, there are 200 elements on the Y-axis. For the X-axis, the number of elements depends on the chosen time resolution and period of the data collection. In this article, while generating the synthetic data, we work with 10-ms total duration and 10-μs resolution, which results in 1000 elements on the X-axis. Therefore, both outputs of the preprocessing steps have the size of $200 \times 1000 \times 1$. The images are then concatenated on the third axis, and the final input image of the network becomes $200 \times 1000 \times 2$.

The preprocessing step iteratively calculates the test statistics for each PRI level, so that it scans the PRI spectrum from 5 to 204 μs. Furthermore, a harmonic PRI is the integer multiple of the true PRI value, and we define the PRI level as the PRI range, in which there cannot be a harmonic PRI. To guarantee that there is not any harmonic PRI in the current iteration, we determine the range of kth PRI level as $[\tau_k, 2\tau_k)$. In this manner, it is easy to see that the first PRI level is [5,10) μs, the second PRI level is [10,20) μs, and so on. To detect the radars with short dwell durations (i.e., dwell switch PRI-type), each PRI level has its own time window length to calculate PRI transform. Time window lengths are set to be proportional to the PRI level, so that radars with short transmission durations are captured easily. Otherwise, they would seem mostly absent in the case of a larger time window and their test statistics would be likely to stay below the threshold, which leads to miss detection. The time window length is decided as $T_k = m\tau_k$, where m is the coefficient that brings the proportionality to the interested PRI level. In our work, m is chosen as 10, which provides enough data collection period for identifying all of the PRI types. Moreover, the time windows are overlapped with the ratio $r = 0.9$ to detect the transient regions better. Therefore, there are $M = (\mathrm{toa}_N - \mathrm{toa}_1)/(T_k \times (1 - r))) \times 2$ PRI transform calculations for a single PRI level in the preprocessing step. The labels generated for the network consist of the following three classes.

1) Class-0 represents the absence in which there is no radar signal transmission at the corresponding PRI level and time duration.
2) Class-I represents constant and dwell switch types, whose PRI values act stable and do not change.
3) Class-II is the class of jittered and staggered types, whose PRI values vary and have a large standard deviation.

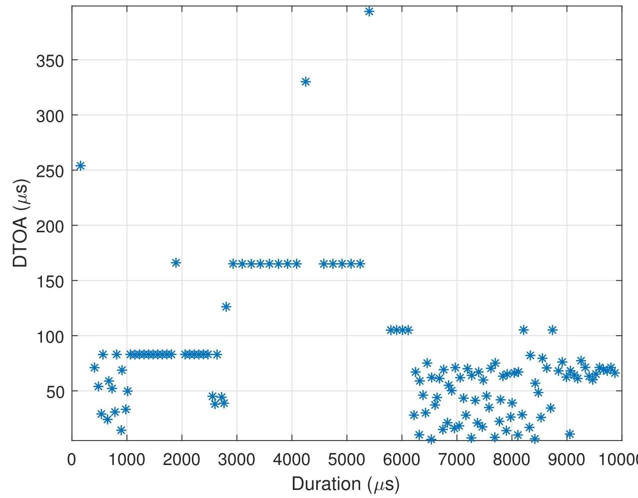

Fig. 2. Difference of TOAs versus duration for the generated data.

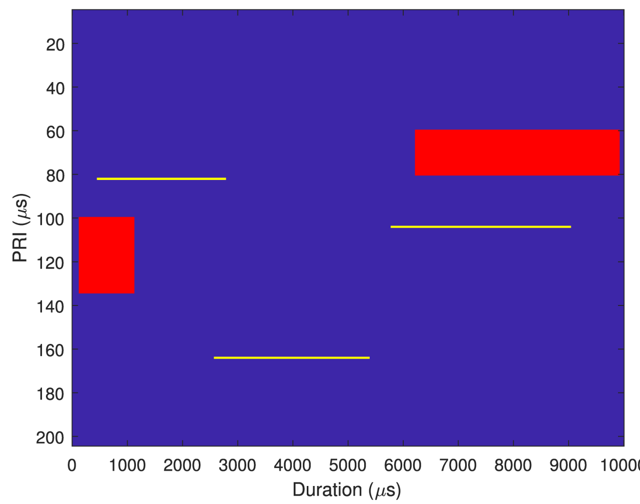

Fig. 3. Label image of five interleaved radar signals, where three of them have class-I PRI-type (yellow) and two of them have class-II PRI-type (red).

Last, the label image is one-hot-encoded, which results in the size of $200 \times 1000 \times 3$.

In Figs. 2 and 3, the DTOA versus time graph and the label image of the five synthetically interleaved radar signals are shown, respectively. It is seen that there are three radars with class-I and two radars with class-II PRI-type. Radars with constant PRI-types have mean PRI values of 82, 103, and 165 μs, while the radars with varying PRI-types have mean PRI values of 70 and 120 μs. Figs. 4 and 5 show the generated images of the proposed preprocessing method. In Fig. 4, it is observed that only constant PRI patterns are detectable, since the PRIT-I is designed for constant PRI-types only. As for Fig. 5, the radar signals with varying PRI patterns appear at the corresponding PRI levels. In addition, the high test statistic for the constant PRI levels is clear in the figure, since PRIT-II can detect both PRI types.

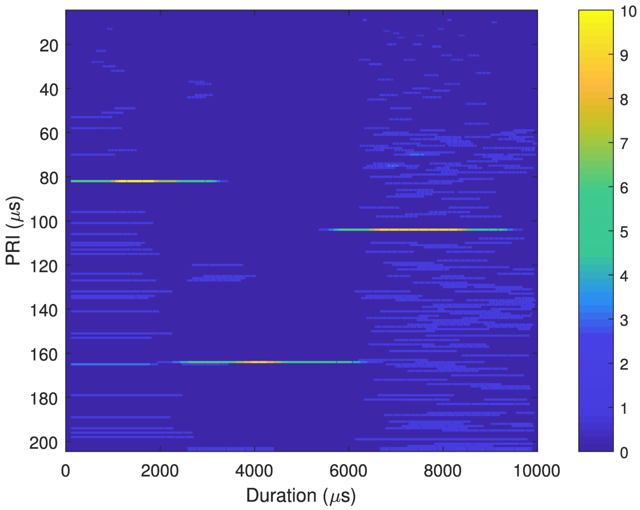

Fig. 4. 2-D input image generated by PRIT-I.

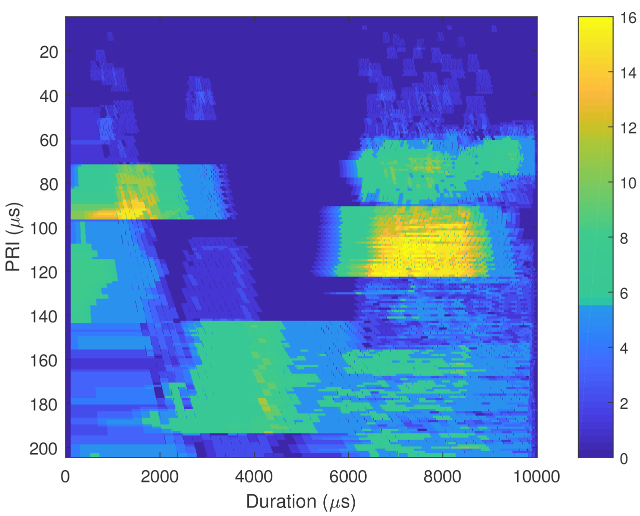

Fig. 5. 2-D input image generated by PRIT-II. Notice that the scale of the coloring is different from Fig. 4.

B. Proposed Network

The proposed network is based on a practicable U-Net structure. Fig. 6 shows the network. It utilizes only 3×3 convolutions and there is no fully connected layer, which results in a fully convolutional network [29]. The number of layers and the sizes of the corresponding hidden outputs are shown on the top and the left-hand side of each layer, respectively. All layers are followed by a rectified linear unit (ReLU) activation function and the batch normalization process, except the last one producing the predicted classes by the softmax activation function [30], [31]. Overall, the network has roughly 130 000 trainable parameters, which is pretty low compared to most of the image segmentation implementations having millions of parameters [28], [32], [33].

C. Simulation Data and Training

To train the proposed network, 50 000 scenarios are created, in which 10% is equally left for validation and testing. Constant, jittered, staggered, and dwell switch PRI-types are considered. More details regarding the simulation parameters are shared [34].

The network is independently trained with three different loss functions: 1) dice; 2) categorical cross-entropy (CCE); and 3) weighted CCE (WCCE). Among them, CCE is more related to accuracy; therefore, it causes instability if the class distribution of the data is not balanced. However, dice and WCCE loss functions inherently handle the unbalanced data, which is in the nature of deinterleaving, i.e., only certain PRI levels and time durations are occupied by the radar signals in the 2-D time–PRI images generated by the proposed method [35]. Let λ_0, λ_1, and λ_2 be the associated weights in the WCCE loss function for classes 0–II, respectively. Then, they are adjusted as $\lambda_1 \gg \lambda_2 > \lambda_0$, so that the network will not favor learning only classes 0 and II, which are present in the label image with very high percent. After fine-tuning, we set $\lambda_0 = 1$, $\lambda_1 = 50$, and $\lambda_2 = 5$. For the optimization procedure, default moment parameters are preferred. The base learning rate is chosen as 10^{-4}. The adaptive moment estimation (ADAM) optimizer for mini-batch gradient descent with a mini-batch size of 32 is used, and the number of training epochs is decided as 20 [36].

Note that the training data are based on only preprocessed TOA measurements. Adding the other pulse description parameters, such as PW, frequency, and AOA, is straightforward but not in the concern of our work.

IV. RESULTS AND ANALYSIS

In this section, first we provide both qualitative and quantitative results on simulation data. Then, we give some qualitative results on real measurements. The real data are collected from a digital narrowband ESM receiver with 100-MHz instantaneous bandwidth and roughly −82-dBm sensitivity level operating in 2–18-GHz frequency band located on a moving platform. The receiver is capable of generating up to one million PDWs per second. We discard PW, frequency, and AOA parameters of the collected PDWs for the analysis of the proposed method. In addition, we do not perform quantitative analysis on the real dataset, since the number of available measurements is not sufficient.

A. Qualitative and Quantitative Results on Simulation Data

Six different scenarios are drawn from the created synthetic dataset. Each scenario corresponds to a unique deinterleaving case with different numbers and types of radars. In Fig. 7, we demonstrate the segmentation results of the proposed method for these scenarios, when the network is trained with the WCCE loss function. Each column represents a single scenario and the first row indicates the label images, while the second row shows the predicted

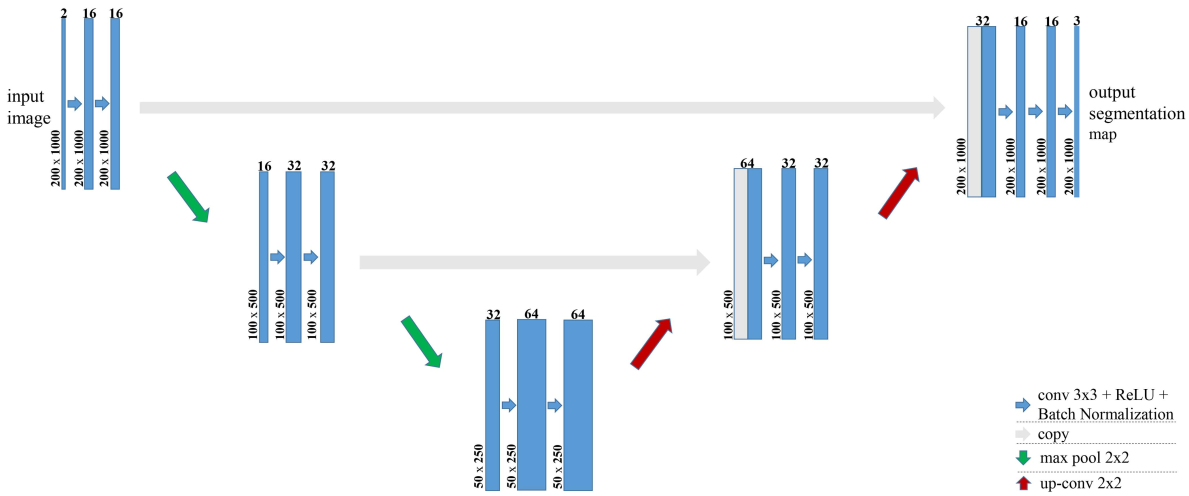

Fig. 6. Proposed network. The number of layers and the sizes of the corresponding hidden outputs are shown on the top and on the left-hand side of each layer, respectively.

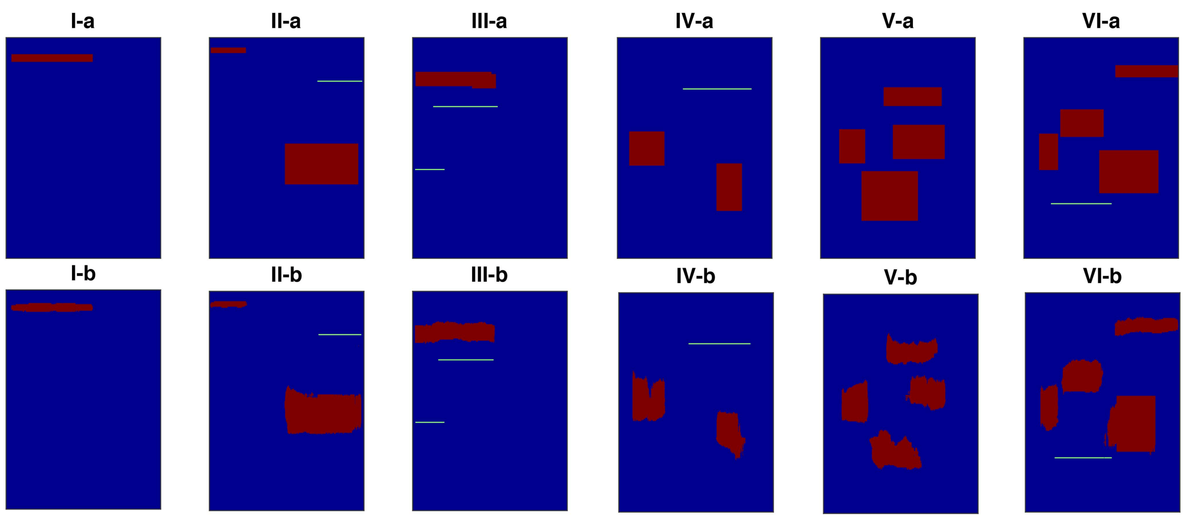

Fig. 7. Segmentation results of the proposed method trained with the WCCE loss function. Axis labels are intentionally removed for better visualization (best viewed once magnified), they are the same as in Figs. 3–5.

images. In the first scenario, there is a single radar with class-II PRI-type, and it is observed that the network is able to produce correct segmentation at the correct PRI level and transmission duration. In the second scenario, there are two radars with class-II PRI-type and a single radar with class-I PRI-type. Notice that two radars are intersected in time. It is seen that the network is capable of successful deinterleaving. For the third scenario, there are two radars with class-I PRI and two radars with class-II PRI-type. The network merges two low PRI levels from class-II, since their centers are very close to each other. This is not a problem and can be easily handled during the pulse-tracking phase. The fourth scenario is similar to the second scenario, and the network seems to separate the intersected two radars as desired. In the fifth scenario, there exist four interleaved radar signals with class-II PRI-type, which is a hard deinterleaving case. The network deinterleaves them successfully in a way that the estimated center PRI levels and transmission durations are almost perfect. Last, the sixth scenario includes five radars, in which four of them have class-II PRI-type and one of them has class-I PRI-type. There are up to three radars simultaneously existing, and the network is observed to be deinterleaving them correctly. Overall, it is seen that the proposed method is more successful at detecting class-I PRI-type than detecting class-II PRI-type, since the radars with class-I PRI-type can be revealed more clearly in the proposed preprocessing method.

To show the performance of the proposed methods objectively, some failure cases are also demonstrated in Fig. 8. Similar to Fig. 7, each column represents a unique scenario and the first row corresponds to the label images, while the second row indicates the predicted images. In the first fail scenario, there are four radars with class-II PRI-type and the EW receiver intercepts them simultaneously. In addition

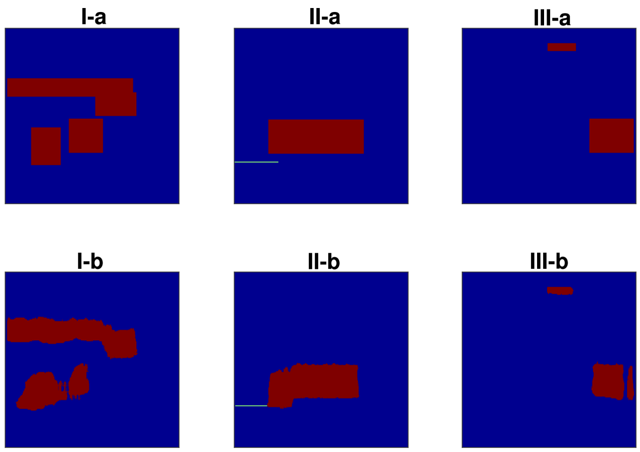

Fig. 8. Failed segmentation results of the proposed method trained with the WCCE loss function (best viewed once magnified).

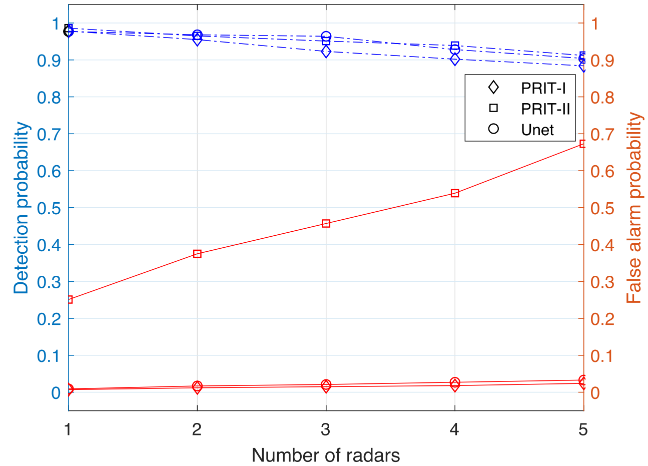

Fig. 9. ROC analysis for the class-I PRI-type. Dashed and continuous lines indicate detection and false alarm probabilities, respectively.

to this, their PRI levels are quite close to each other such that the red rectangles overlap. Here, the proposed method fails to determine the PRI level borders exactly. In the next scenario, there are two radars with different PRI types. The proposed method inaccurately expands the region of the radar with varying PRI. In the last scenario, there are two radars with class-II PRI-type. Different from the previous failure cases, the proposed method does not merge the PRI regions of the radars, since their PRI levels are far away. Instead, it splits one of the regions into two.

To quantitatively demonstrate the enhancement that is brought by the proposed method, outputs of the PRI transform methods are binarized by the threshold defined in (7), so that the signal absence and presence are labeled as 0 and 1, respectively. Moreover, the output of the proposed network is also binarized by labeling the predicted class-0 PRI-type as signal absence, classes I and II PRI-types as signal presence. Then, PRIT-I, PRIT-II, and the proposed model trained with the loss functions (i.e., dice, CCE, and WCCE) mentioned in Section III-C are compared according to the segmentation metrics: accuracy, Jaccard index, and SSIM. Here, accuracy is defined as the average pixelwise accuracy, while Jaccard index is a 1×2 vector indicating intersection over union values for signal absence and presence. For the null hypothesis $\mathcal{H}_0$ and alternate hypothesis $\mathcal{H}_1$, it is calculated as

$$J(\mathcal{H}_0) = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FN}+\mathrm{FP}}$$
$$J(\mathcal{H}_1) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{8}$$

where TP, FP, and FN represent the true positive, false positive, and false negative records, respectively. Moreover, SSIM is defined as the multiplication of three similarity terms: 1) the luminance term; 2) the contrast term; and 3) the structural term [37]. Different from the other metrics we preferred, it focuses on SSIMs of the image patches rather than individual pixels. It was initially designed for the evaluation of compression or transmission artifacts. However, lately, some works inherit SSIM as a loss function and a performance metric for the deep image segmentation networks [38], [39]. It is calculated as

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{9}$$

where μ_x, μ_y, σ_x, σ_y, and σ_{xy} are the local means, standard deviations, and cross covariance for images x and y, while C_1 and C_2 are the regularization constraints to avoid instability for image regions with zero local mean and standard deviation. Here, the constants C_1 and C_2 are taken as $(0.01\,L)^2$ and $(0.03\,L)^2$, with $L = 255$ denotes the dynamic range.

Table I gives the segmentation performances for $\{0, 10, 20\}$ percent missing pulse cases, which is a typical simulation scenario for ESM applications [34]. The best results are indicated in bold. It is observed that PRIT-I and PRIT-II are outperformed by the proposed method in all missing pulse ratio scenarios. The accuracy of the PRIT-I is lower than that of PRIT-II, since it cannot capture varying PRI-types. However, its Jaccard index for the null hypothesis is higher than that of PRIT-II, as PRIT-II is likely to spread the peak of the test statistics of the constant PRI patterns into several PRI levels that are adjacent to the center PRI. Moreover, the proposed method reaches the best Jaccard and SSIM indices when the loss function is chosen as WCCE with the suggested hyperparameters. In addition, it is observed that the dice loss function results in better Jaccard and SSIM indices than CCE, while it has insignificantly lower accuracy.

In Fig. 9, a receiver operating characteristic (ROC) analysis of the scenarios, including the radars with only class-I PRI-type, is shared, in which the proposed method is trained with WCCE loss function. We calculate the detection and false alarm probabilities pixelwise in the same manner as the previous results, while the number of the radars varies on the X-axis. All methods have a high detection probability regardless of the number of radars. However, the performances differ in false alarm probability. PRIT-II has the highest false alarm, since the constant PRI-type causes its PRI spectrum to be spread around the center

TABLE I
Segmentation Performance: [Accuracy | Jaccard Index ($\mathcal{H}_0$, $\mathcal{H}_1$) | SSIM] Values for Different Missing Pulse Percentage Cases

%	PRIT-I	PRIT-II	Dice	CCE	WCCE
0	0.717|(0.917, 0.104)|0.422	0.902|(0.838, 0.180)|0.571	0.950|(0.946, 0.716)|0.871	**0.956**|(0.953, 0.682)|0.865	0.937|(**0.984**, **0.812**)|**0.884**
10	0.709|(0.922, 0.103)|0.387	0.812|(0.808, 0.161)|0.509	0.912|(0.926, 0.676)|**0.846**	0.922|(0.915, 0.659)|0.831	**0.924**|(**0.944**, **0.795**)|0.818
20	0.653|(0.858, 0.095)|0.324	0.723|(0.759, 0.131)|0.443	0.837|(0.881, 0.634)|0.813	**0.898**|(0.871, 0.611)|0.787	0.895|(**0.919**, **0.768**)|**0.815**

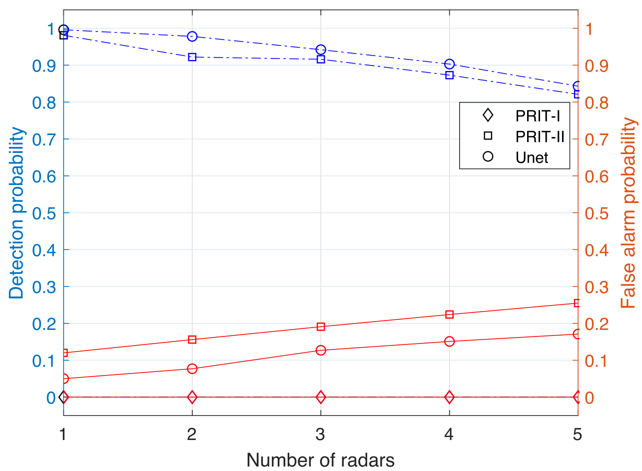

Fig. 10. ROC analysis for the class-II PRI-type. Dashed and continuous lines indicate detection and false alarm probabilities, respectively.

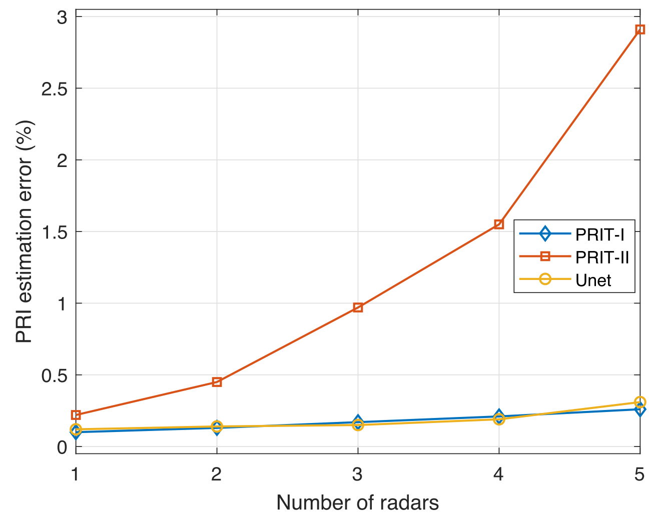

Fig. 11. PRI estimation error (percentages) for class-I PRI-type.

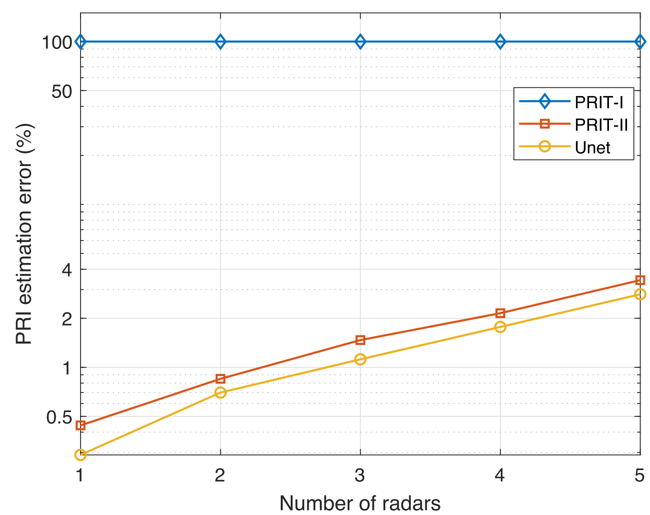

Fig. 12. PRI estimation error (percentages) for class-II PRI-type. Note that y-axis is logarithmic.

PRI level. PRIT-I yields almost zero false alarm ratio, since it is designed for only constant PRI-type. In addition, the proposed method performs quite similar to PRIT-I, since it fuses both information coming from PRIT-I and PRIT-II preprocessing images.

In Fig. 10, the ROC analysis of the scenarios, including the radars with only class-II PRI-type, is investigated with the same configuration in Fig. 9. PRIT-I is unable to make detections, since the PRI values vary. This causes zero detection and false alarm probability. Moreover, the proposed method outperforms PRIT-II in both detection and false alarm probabilities, which means better pixelwise accuracy is provided regardless of the number of existing radars. We also observe that its performance degrades insignificantly as the number of radars increases.

The metrics that we utilize so far analyze the image segmentation performance of the deinterleaving algorithm. To investigate the pulse sorting performance of the proposed method, we accompany the last metric that has been used by several deinterleaving papers [40]–[42]. It is the average $\mathcal{L}_1$ error between the estimated PRI levels and the true PRI level. It is calculated as

$$p\text{Error} = \frac{1}{N}\sum_{i=1}^{N}\frac{\|\hat{\tau}_i - \tau_i\|}{\tau_i} \tag{10}$$

where N is the number of existing radars, $\hat{\tau}_i$ is the estimated PRI level, and τ_i is the true PRI level. If the method could not detect an existing PRI level, $\hat{\tau}_i$ is set to 0 so that the maximum contribution to the error value is provided. Each PRI level is estimated as the mean PRI value of the pixels coming from the same cluster of the segmentation output. Figs. 11 and 12 show the PRI estimation error for classes I and II PRI-types in percentages, respectively. The same configuration in Figs. 9 and 10 is used and the number of the radars varies on the X-axis. We observe that the proposed method achieves almost identical results with PRIT-I for class-I PRI-type, while PRIT-II performs worse with error percentages below 5. Furthermore, PRIT-I is unable to detect class-II PRI-type, which causes high estimation error

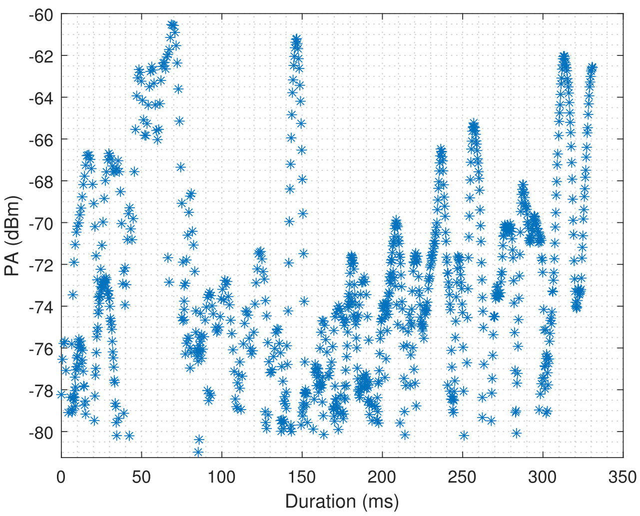

Fig. 13. PA of three interleaved radar pulses.

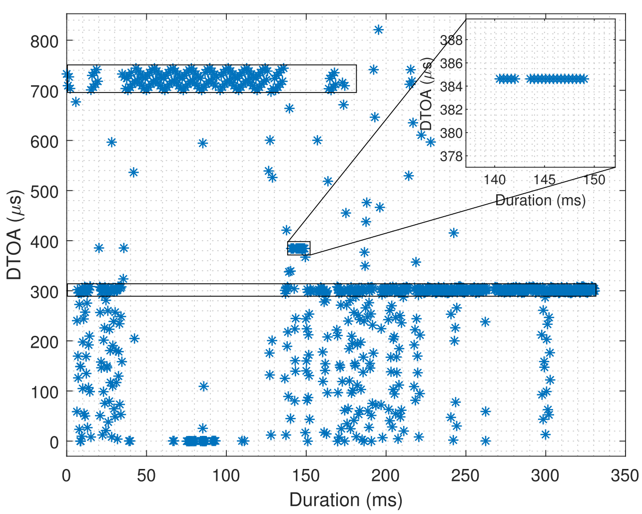

Fig. 14. DTOA of three interleaved radar pulses. Each rectangle indicates the pulses from a unique emitter.

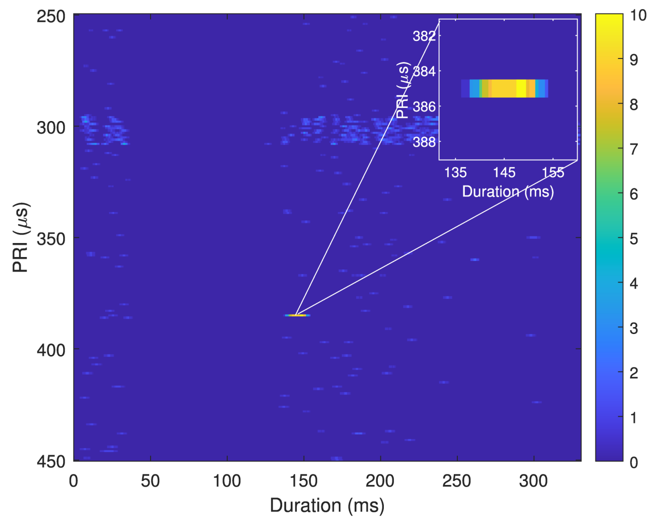

Fig. 15. 2-D input image generated by PRIT-I.

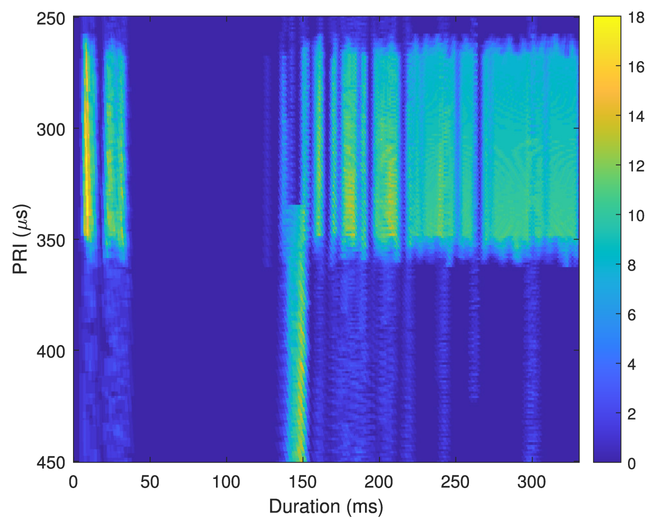

Fig. 16. 2-D input image generated by PRIT-II.

percentages at 100, and the proposed method provides similar performance to PRIT-II's, while being slightly better. Overall, the PRI estimation error analysis matches the image segmentation performance analysis that we make earlier.

B. Qualitative Experiments on Real Data

Two different measurements, which include multiple radar signals simultaneously, are extensively investigated in terms of both preprocessing and deinterleaving quality.

In Fig. 13, the pulse amplitude (PA) of three interleaved radar signals is shown. Due to the scanning-type of the radars, their amplitudes are seen in the shape of the sinc function [43]. DTOA of these interleaved pulses is shown in Fig. 14. Several small DTOA values are observed in addition to the real PRI values of the radars, which result from the interleaving of the received radar signals. Note that the largest DTOA value in a certain duration of the graph is upper bounded by the radar with the smallest PRI that is active in that duration. Furthermore, the receiver intercepts two of three radars having 300- and 725-μs center PRI values for a long duration. The other radar having 385-μs center PRI value with constant-type is observed for a very short duration, which is less than 20 ms. On the right-top of the figure, DTOA values and the duration of the transmission are particularly shown for the last radar.

To obtain the output of the proposed neural model, first 2-D time–PRI input images are generated by the developed preprocessing method. The preprocessing method is applied to the whole PRI range. However, in this article, only the PRI bounds, which do not generate a null hypothesis, are analyzed. The first PRI boundary is [250,449] μs that captures the radars with 300- and 385-μs center PRI values, and the second PRI boundary is [650,849] μs covering the radar with 725-μs center PRI value.

Figs. 15 and 16 show the two outputs of the preprocessing method that are applied on the PRI levels $\in$ [250, 449] μs. Since only radars with class-I PRI-type can be detected by PRIT-I, the radar with 385-μs center PRI value and constant PRI-type yields high test statistics on the image

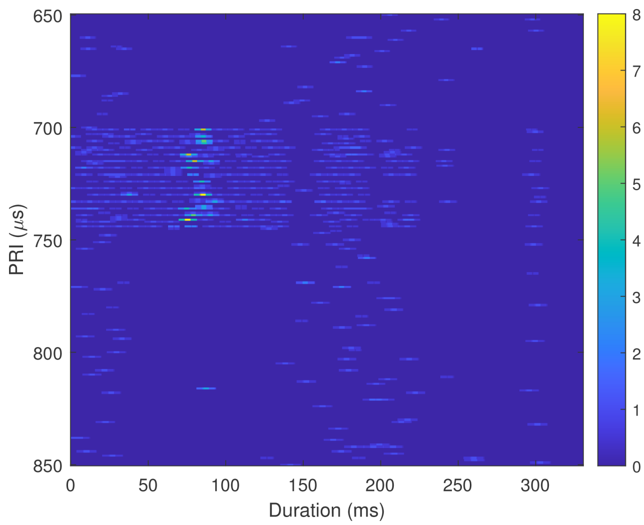

Fig. 17. 2-D input image generated by PRIT-I.

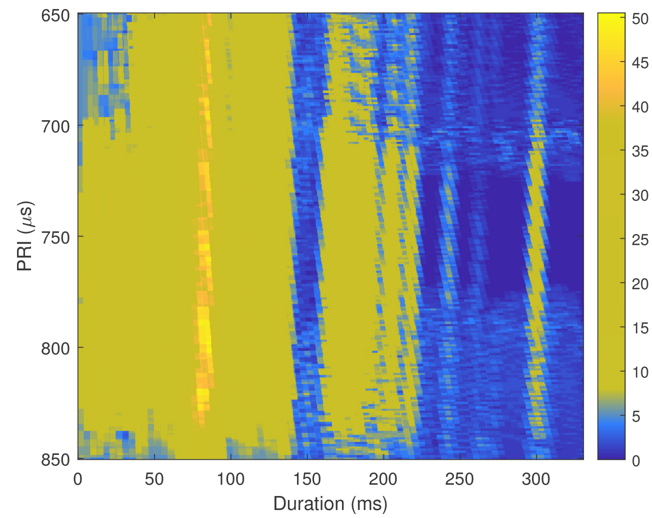

Fig. 18. 2-D input image generated by PRIT-II.

generated by PRIT-I, which is shown in Fig. 15. The radars with 300- and 725-μs center PRI values correspond to jittered and staggered PRI-types, and they belong to class-II PRI-types. Therefore, their test statistics on the figure are mostly zero. Detection of the radar with 300-μs center PRI value is enabled by the image generated by PRIT-II, which is shown in Fig. 16. As observed, both radars with 300- and 385-μs center PRI values return high test statistics. Notice that the radar with 725-μs center PRI value is out of borders [250,449] μs in Figs. 15 and 16, and therefore cannot be observed.

The radar with 725-μs center PRI value is observed when the preprocessing method is applied on the PRI levels $\in [650, 849]$ μs. Fig. 17 shows the image generated by PRIT-I. There are high test statistics for PRI levels $\in [700, 740]$ μs at duration $\in [70, 85]$ ms even though there is no radar transmission with that PRI level at that duration. Since the receiver intercepts the radar signal in its sensitivity level, the receiver splits most of the pulses into smaller durations. Therefore, their PW and DTOA levels are much lower than their actual values. This causes more pulses to fall into the PRI bin that the PRI transform operation is applied, which results in a much higher test statistics value. More information regarding this pulse splitting phenomenon are shared [44]. In Fig. 18, the image generated by PRIT-II is shown, where the existence of radar with 725-μs center PRI value can be noticed. The test statistics have much higher values compared to Fig. 16, since the PRI values with 725-μs center level are in the range of the first harmonic of the PRI values with 300-μs center level. Notice that the aforementioned phenomenon can also be seen in the figure. The test statistics have higher values in [70,85]ms duration due to the pulse splitting behavior of the receiver.

The three radars are revealed through the proposed preprocessing technique by forming 2-D time–PRI images that are concatenated and then fed to the neural network. The segmentation result, which is taken from the output of the model trained with WCCE loss function, for PRI boundary [250,449] μs, is shown in Fig. 19. The radar with 385-μs center PRI value of class-I type is detected and classified correctly in a way that its existence on the figure has the same position as its position on Fig. 15, and its color is yellow indicating its constant PRI-type. The second radar that falls into the current PRI boundary is the radar with 300-μs center PRI value of class-II type, which is also detected and classified correctly, such that its existence on the image lies in the PRI boundary [275,325] μs, and its color is red indicating its varying PRI-type. Moreover, Fig. 20 shows the segmentation result for the PRI boundary [650,849] μs, and it is observed that the radar with 725-μs center PRI value of class-II type is correctly detected and classified. The center PRI value of the red cluster in the image is almost equal to the true center level and its PRI type is classified as Class-II, while the detection region is in good agreement with Fig. 14.

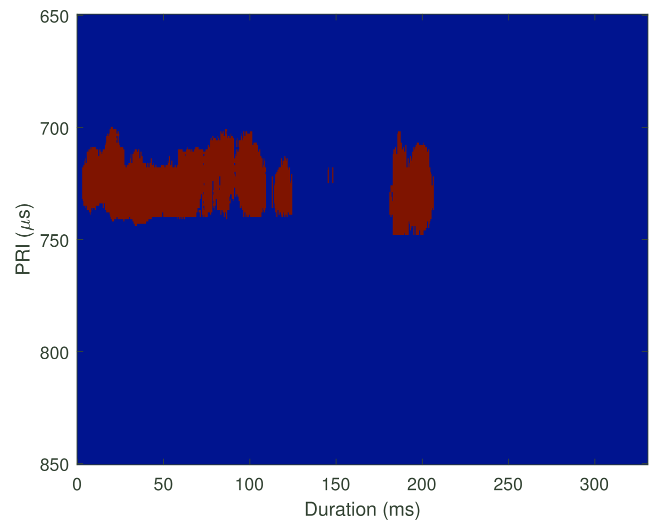

Fig. 19. Segmentation results of the proposed method trained with WCCE loss function for the PRI boundary [250,450] μs.

In Figs. 21 and 22, PA and DTOA parameters of another real measurement are shown. Three radar signals exist in the

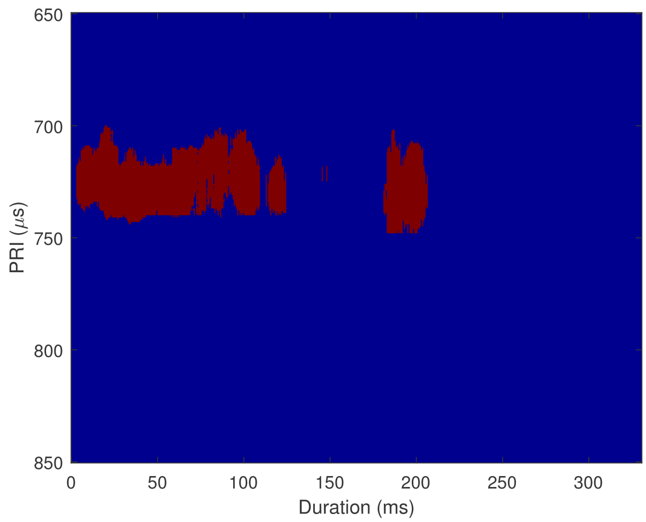

Fig. 20. Segmentation results of the proposed method trained with WCCE loss function for the PRI boundary [650,850] μs.

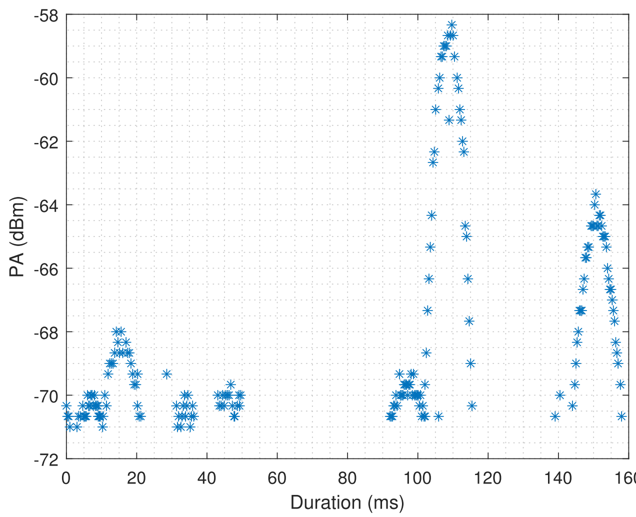

Fig. 21. PA of three interleaved radar pulses.

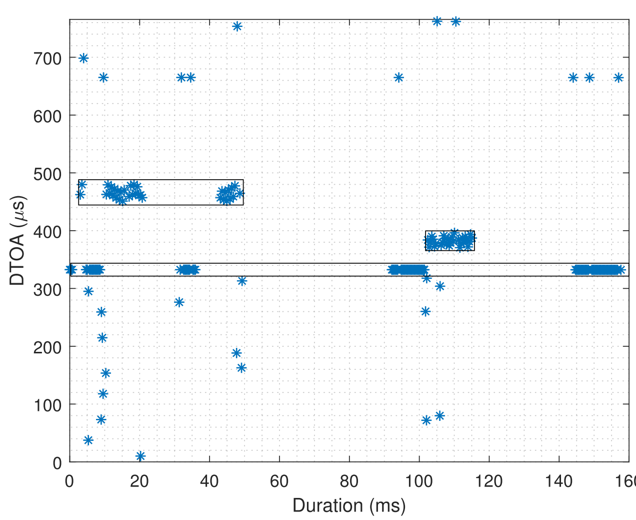

Fig. 22. DTOA of three interleaved radar pulses. Each rectangle indicates the pulses from a unique emitter.

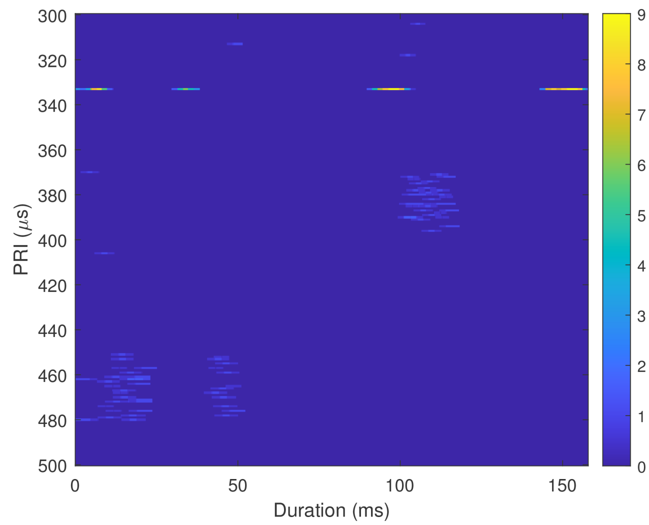

Fig. 23. 2-D input image generated by PRIT-I.

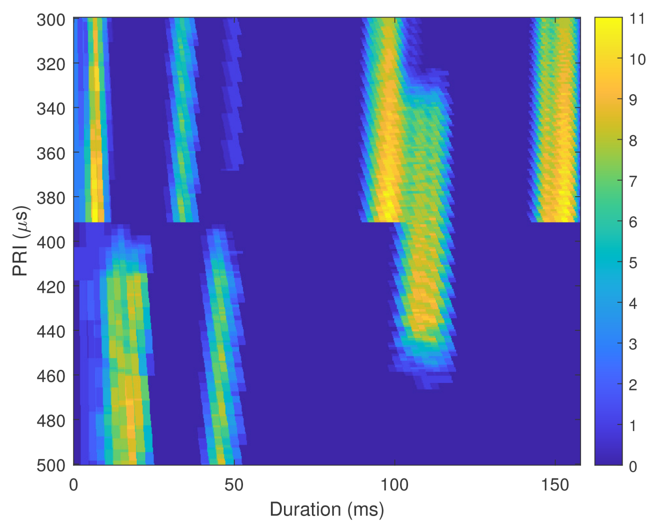

Fig. 24. 2-D input image generated by PRIT-II.

receiving duration: 1) the first radar has staggered PRI with 460-μs center value; 2) the second radar has constant PRI centering around 333 μs; and 3) the last radar has jittered PRI with 385-μs center value. PRI transform is applied to the whole PRI range. Different from the experiment on the first real measurement, a single PRI boundary, which captures all PRI levels of the existing radars, is sufficient to analyze. The 2-D time–PRI input images corresponding to the preprocessing of the PRI boundary [300,499] μs, which can be seen in Figs. 23 and 24, are fed to the model trained with WCCE loss function. The radar with constant PRI-type is shown in Fig. 23, where the regions of high test statistics on the image are in parallel with Fig. 22. Since the other two radars have class-II PRI-types, they are detected in the image generated by PRIT-II, which is shown in Fig. 24. Both radars with class-II PRI yield high test statistics on the image in a way that there are two major regions with center PRI values of 333 and 460 μs. Notice that their discontinuity on the time axis of the image can also be verified by their duration of existence in Fig. 22. Moreover, the radar with

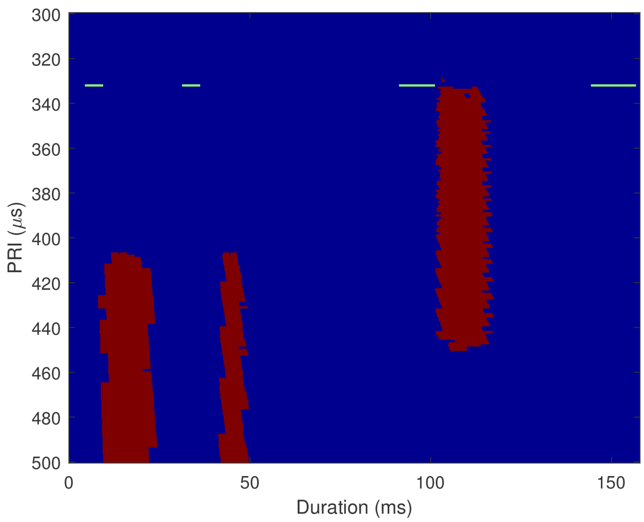

Fig. 25. Segmentation results of the proposed method trained with WCCE loss function for the PRI boundary [300,500] μs.

constant PRI-type also causes high test statistics on the image, and its power in the PRI spectrum is spread around its center value, which is the same result obtained from the synthetic data case.

The segmentation result of the data for PRI boundary [300,499] μs is shown in Fig. 25. The radar with 333-μs center PRI value of class-I type is detected, and its color is yellow indicating that the classification is successful. The radars with class-II PRI-types can be observed at the right duration and right PRI levels in a way that the red regions are centered around 385- and 460-μs PRI values.

Radars manipulate and set their RF, PW, and PRI parameters depending on their mode of operation. Since the EW system does not priorly know the frequency band of the radar, it has to scan the spectrum. Moreover, both EW and radar systems scan the space by using rotating antennas or electronically scanning antennas. In addition, most of the pulsed radars have a pretty low duty cycle, and their beamwidths are very narrow. All of these settings cause the interception to occur with a very low probability. Therefore, interception of multiple simultaneous radars gets more and more unlikely, as the number of radars increases. Thus, we can say that the deinterleaving of the real measurements that we share in this article is not a typical scenario for the EW system, considering that there are up to three simultaneous radars in these measurements. Furthermore, a synthetically generated scenario consists of up to five simultaneous radars, which we believe is sufficient for narrowband receivers. Our receiver is narrowband with 100-MHz instantaneous bandwidth, which is far lower than the total bandwidth of 2–18 GHz. As the instantaneous bandwidth of the receiver widens, the probability of interception of more simultaneous radars increases. In that case, the simulation data might be expanded in a way that it can deal with more simultaneous radars. The proposed method effectively works on the PDWs provided by our receiver. Therefore, it can work on other narrowband receivers as well.

TABLE II
Execution Time Analysis: Average Time Consumption (in Seconds) in Different Scenarios for the PRIT-I, PRIT-II, Proposed Preprocessing Technique, and Proposed Deep Model

Case	PRIT-I	PRIT-II	Prepr.	Infer.
I-a	0.10	0.14	0.21	0.06
II-a	0.14	0.19	0.29	0.07
III-a	0.21	0.31	0.46	0.07
I-b	0.10	0.18	0.24	0.06
II-b	0.13	0.24	0.32	0.07
III-b	0.22	0.39	0.46	0.08

C. Complexity Analysis

The PRI transform operation has the complexity of $\mathcal{O}(N^2)$, where N indicates the number of pulses. Therefore, both PRIT-I and PRIT-II have $\mathcal{O}(N^2)$ complexity, while PRIT-II requires more computation due to time origin shifts and harmonic suppression. Our method applies both methods and generates the input images. Since the proposed deep neural network is fully convolutional, it only filters the input images by the convolution operation. It is a fast operation, as it only requires summations and multiplications [29]. In addition, the network is intentionally designed to be small, so that it yields fast inference. Table II gives the average time consumption (in seconds) of these three methods. The methods run on an Intel i7-8750H CPU working at 2.20 GHz, and the same synthetic scenarios described in Section III-C are utilized. In the table, each row indicates a different simulation case. Cases I-a, II-a, and III-a correspond to the scenarios, in which one, three, and five simultaneous radars with class-I PRI-type exist, respectively. Similarly, cases I-b, II-b, and III-b indicate the scenarios of the same number of simultaneous radars with class-II PRI-type. Moreover, there are four information columns in the table. To comprehend the complexity of the proposed method part by part, execution time analysis is separated into two: 1) preprocessing (Prepr.); and 2) inference (Infer.). They show the average consumed time by the proposed preprocessing technique and the deep model, respectively.

As the number of radars increases, both PRI transform methods require more duration to be completed. The developed preprocessing technique uses both of these methods, except the calculation of the threshold and its application process. Hence, its average time consumption is smaller than the sum of their individual values. Furthermore, the total number of the complex autocorrelation calculations does not differ in both transform methods whether the PRI is constant or varying. However, PRIT-II performs slower if the PRI values are varying, while PRIT-I performs quite similarly. Since the PRI bins are wider in PRIT-II than that of PRIT-I and they are overlapped; a single complex autocorrelation value matches more PRI levels. This causes more summations and slow performance for PRIT-II if the PRI varies. As for the proposed method, the main complexity comes from the preprocessing step and the average inference duration changes insignificantly as the number of radars increases. Overall, the proposed method has higher complexity than the compared methods, since it utilizes both of them in the preprocessing step in addition to deep neural inference.

V. CONCLUSION

In this article, we approached the deinterleaving problem from the image segmentation perspective. A novel preprocessing method was proposed to generate images from the raw TOA data. The images were then pixelwise classified by the proposed deep neural network, which was based on a practicable U-Net structure. The network was independently trained with dice, CCE, and WCCE loss functions to observe the effect of the data imbalance, and it was compared to PRIT-I and PRIT-II methods. In addition to qualitative and quantitative experiments on synthetic data, we conducted qualitative experiments on real data and shared the results. We showed that the proposed method utilizes both PRI transform methods successfully in the preprocessing step, so that it revealed different PRI types in the generated 2-D time–PRI images. Furthermore, the proposed method outperformed PRIT-I and PRIT-II in terms of segmentation accuracy, Jaccard index, SSIM, and PRI estimation error metrics. It is also observed that the model trained with WCCE loss function and suggested hyperparameters yields the best performance compared to the other loss functions.

Since the frequency spectrum of radar and communication signals intersect, the EW receiver intercepts both types of signals. To make our work more generalizing, in the next phase of this work, synthetic communication signals will be created and added to the existing training dataset, so that further results regarding the process of both signals will be obtained. Another possible future work could be to develop a new preprocessing method and deep learning model to accompany more PDWs parameters, including PW, frequency, and AOA.

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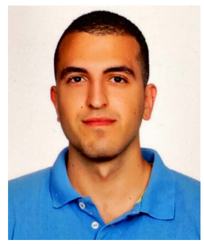

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